

Revenue-Sharing Coordination of Water Distribution and Distributed Energy Operators Under Network Constraints

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Abstract—Municipal water pumping is a storage-backed electrical load whose flexibility is worth most to whoever holds the site's day-ahead market position, and that is increasingly a different firm: a local energy operator with photovoltaic generation, a battery, and the grid connection. This paper treats the day-ahead coordination of the two operators as a contracting problem while keeping the physics of both networks. The water operator's decisions are constrained by the MILPNet formulation of hydraulics, tanks, pumps, and valves; the energy operator's by the electrical part of the distributed-energy-system model of Morvaj *et al.*, with linearized AC power flow and branch-current limits. The two blocks meet in one metered quantity, the pump load at its supply bus, and their joint optimum is a mixed-integer linear program. We show that regulated tariffs leave the resulting coordination gain uncaptured or captured by accident, and that a contract under which the energy operator pays the water operator a share of its day-ahead settlement plus a fixed fee makes the jointly optimal schedule a Nash equilibrium at every share, with the fee dividing the gain in any proportion; because the site is a net importer the contract reads as bill sharing. The schedule is computed without pooling data: a Lagrangian relaxation of the interface lets the operators exchange only an internal price and a metered load schedule, and a bundle-type master problem certifies a dual bound in a few dozen exchanges despite the integer pump commitments. The internal price equals the purchase price in import hours, the sale price in export hours, and departs from both only where the connection or the feeder is at a limit. In a case study with a small municipal network and a 100 kW feeder with a 200 kW_p array, coordination is worth 17.03 per day, 46% of the water operator's pumping bill; the pass-through tariff captures 95.6% of it, and the contract captures all of it at every share. Replaying the schedules in EPANET and OpenDSS confirms pressure and voltage limits at report times.

Index Terms—Water-energy nexus, day-ahead market, revenue-sharing contract, Lagrangian relaxation, distributed optimization, mixed-integer linear programming, pump scheduling, water distribution systems.

NOMENCLATURE

A. Sets and Indices

$\mathcal{T}, \bar{\mathcal{T}}$	Operating intervals $\{0, \dots, T-1\}$ and inventory boundaries $\{0, \dots, T\}$, index t .
W, E	Local water operator; local energy operator.
$\mathcal{J}, \mathcal{K}, \mathcal{R}$	Water junctions, tanks, and reservoirs; $\mathcal{N} = \mathcal{J} \cup \mathcal{K} \cup \mathcal{R}$, index n (or i, j for link ends).
$\mathcal{L}^P, \mathcal{P}, \mathcal{V}$	Pipes, pumps, and valves; $\mathcal{L} = \mathcal{L}^P \cup \mathcal{P} \cup \mathcal{V}$, index l (p for pumps, v for valves).

$\mathcal{B}, \mathcal{E}, \mathcal{G}$	Electrical buses, lines, and grid-connection buses $\mathcal{G} \subseteq \mathcal{B}$; indices b and l .
$\mathcal{P}_b, \mathcal{B}_P$	Pumps supplied from bus b ; buses that supply a pump.
A_{nl}^w, A_{bl}^e	Node-link incidence (+1 entering, -1 leaving, 0 otherwise) of the water and electrical graphs.

B. Parameters

Δt	Interval length (h); Δt_s the same in seconds.
$d_{n,t}^w$	Water demand at junction n (m ³ /s).
z_n, π_n	Junction elevation and minimum pressure head (m).
$H_{r,t}$	Reservoir total head (m).
K_l	Hazen-Williams resistance coefficient of pipe l .
A_k, h_k^0	Tank cross-sectional area (m ²) and initial head (m).
$h_k, \bar{h}_k$	Tank head bounds (m).
a_p, b_p, ν_p	Pump head-curve coefficients, $g_p(q) = a_p - b_p q^{\nu_p}$.
$\bar{q}_l, M_p$	Flow bound of link l (m ³ /s) and pump disjunction constant (m).
$d_{b,t}^P, d_{b,t}^Q$	Non-pump active and reactive load at bus b (kW, kvar).
$\bar{p}_{b,t}^{pv}$	Available photovoltaic power (kW).
$\bar{e}_b, e_b^0$	Battery energy capacity and initial energy (kWh).
$\bar{p}_b^{ch}, \bar{p}_b^{dis}$	Battery charge and discharge power limits (kW).
$\eta_b^{ch}, \eta_b^{dis}$	Battery charge and discharge efficiencies.
$\bar{p}^g, \bar{p}^x$	Connection import and export ratings (kW).
R_l, X_l	Line series resistance and reactance (p.u.).
G_l, B_l	Line series conductance and susceptance (p.u.).
$\bar{i}_l$	Line current limit (p.u.).
$\underline{v}, \bar{v}$	Voltage magnitude bounds (p.u.); S_B power base (kVA).
c_t^{buy}, c_t^{sell}	Day-ahead purchase and sale prices (unit/kWh).
$\bar{\tau}, \tau_t$	Flat and time-varying retail tariffs charged to W (unit/kWh).
$\bar{P}_p, \eta_p$	Calibrated on-state electrical power (kW) and wire-to-water efficiency of pump p .
φ, F	Contract: share of the settlement; fixed fee (unit/day).

C. Decision Variables

$q_{l,t}, h_{n,t}$	Link flow (m ³ /s) and nodal total head (m).
$\Delta h_{l,t}$	Pipe head loss (m).

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Code, network inputs, and the experiment scripts that regenerate the numerical section of this paper are available in the accompanying repository.

$u_{p,t}$	Pump status, 1 when operating (binary).
$\gamma_{p,t}$	Pump head gain (m).
$y_{g,t}$	Gate-valve open status (binary).
$\alpha_{v,t}, \omega_{v,t}, \kappa_{v,t}$	PRV active, open, and closed status (binary).
λ, δ	Piecewise-linear weights (continuous) and segment selectors (binary).
$p_{b,t}^g, p_{b,t}^x$	Grid import and export (kW); $y_{b,t}^g, y_{b,t}^x$ their binary modes.
$r_{b,t}^g$	Reactive power drawn from the grid (kvar).
$p_{b,t}^{pv}$	Dispatched photovoltaic power (kW).
$p_{b,t}^{ch}, p_{b,t}^{dis}$	Battery charge and discharge at the AC bus (kW); $y_{b,t}^{ch}, y_{b,t}^{dis}$ their binary modes.
$e_{b,t}$	Battery energy at boundary t (kWh).
$\ell_{b,t}$	Metered pump load at bus b (kW); $w_{b,t}$ the load E accepts in its subproblem.
$f_{l,t}^P, f_{l,t}^Q$	Line active and reactive flow in the orientation of line l (kW, kvar).
$v_{b,t}, \theta_{b,t}$	Bus voltage magnitude (p.u.) and angle (rad).
$i_{l,t}^{re}, i_{l,t}^{im}$	Real and imaginary line-current components (p.u.).
$\phi_{l,t}, \chi_{l,t}$	Absolute values of $i_{l,t}^{re}$ and $i_{l,t}^{im}$; $\hat{\phi}_{l,t}, \hat{\chi}_{l,t}$ their piecewise-linear squares.
$\pi_{b,t}$	Internal (coordination) price at bus b (unit/kWh).

D. Derived Quantities

R	E's day-ahead settlement, $\Delta t \sum_t \sum_{b \in G} (c_t^{sell} p_{b,t}^x - c_t^{buy} p_{b,t}^g)$ (unit/day).
Π, Π^*, Π^0	Joint payoff; its integrated optimum; its base-line value.
$\mathcal{G}$	Coordination gain $\Pi^* - \Pi^0$ (unit/day).
$T(\varphi, F)$	Contract transfer from E to W (unit/day).
$D(\pi)$	Dual function of the relaxed interface.

I. INTRODUCTION

WATER distribution systems are among the largest electricity consumers operated by municipalities, and pumping accounts for most of that consumption [1], [2]. Because elevated storage decouples the time at which water is pumped from the time at which it is consumed, pumping is a shiftable load: tanks can be filled before expensive hours and drawn down afterwards, and the same storage makes water utilities natural participants in day-ahead scheduling and demand response [3], [4]. As wholesale prices become more volatile with rising renewable penetration, that flexibility grows more valuable.

An organizational obstacle stands between the flexibility and its value. The water utility's mandate and expertise are hydraulic: meet demand, maintain pressure and storage, protect water quality. The electricity market interface increasingly belongs to a different firm, a local energy operator such as a community energy company, campus utility, or microgrid operator, that owns behind-the-meter photovoltaic (PV) generation and battery storage and participates in the day-ahead

market through a single point of common coupling [5], [6]. The two capabilities are complementary but separately held: the schedulable load and its storage sit with the water utility, while the market interface, the forecasting capability, and the settlement infrastructure sit with the energy operator. Neither can realize the value of water flexibility alone.

A companion two-agent model [7] showed, for a pump station with a single storage volume and an energy operator with a market position, that regulated tariffs cannot coordinate the two firms, that a day-ahead revenue-sharing contract does, and that a dual decomposition computes the jointly optimal schedule while the operators exchange only a price signal and a metered load. That analysis rests on a continuous pump, a linear program (LP), and a lumped connection. Real water systems have on/off pumps, pipe friction, minimum service pressures, and tank bounds; real feeders have voltage and thermal limits and several buses. The present paper asks whether the contracting result and the limited-information implementation survive when both operators are described by validated network models: the MILPNet water-network framework of Thomas and Sela [8] and the electrical part of the distributed-energy-system (DES) model of Morvaj *et al.* [9]. The joint problem is then a mixed-integer linear program (MILP) rather than an LP, energy is no longer neutral across schedules, the internal price becomes nodal, and the Lagrangian dual acquires a duality gap.

The contributions are as follows.

- 1) **A two-operator model with network physics.** The water operator's private set is the MILPNet block (nodal mass balance, piecewise-linear Hazen–Williams losses, tanks, pump and valve logic), the energy operator's the DES block (hub balance, storage, linearized AC power flow, branch-current limits). They share exactly one metered quantity, the pump load at its supply bus, and their joint optimum is a MILP whose constraints are mapped to their published sources (Secs. IV–VI).
- 2) **Why tariffs cannot do the job.** A flat tariff selects only the number of pump-hours and leaves their timing to operating practice; a pass-through tariff steers the water operator to the market price but not to the site's internal price, which differs from it in export, idle, and congested hours; even the internal price leaves ties among equally priced hours and fixes the division of the gain as a side effect (Sec. VII).
- 3) **A revenue-sharing contract that coordinates.** The energy operator pays the water operator a share of its day-ahead settlement plus a fixed fee. Following the jointly optimal schedule is a Nash equilibrium at every share, whatever the pump physics, and anchoring the fee on the baseline bill divides the gain in any proportion; for a net-importing site the contract is bill sharing (Sec. VIII).
- 4) **An implementation that respects the boundary.** A Lagrangian relaxation of the interface separates the MILP into the water operator's ordinary price-based scheduling problem and the energy operator's dispatch problem with a price-responsive load. A bundle-type master problem run by the energy operator certifies

a dual bound with only prices and metered schedules crossing the boundary; the internal price is characterized as the nodal marginal cost of serving the pump (Sec. IX).

- 5) **A quantified case study with replay.** A 24-hour study with a small municipal network and a two-bus feeder with PV, battery, and connection ratings measures the gain, the tariff benchmarks, the payoffs and fee window of the contract, the convergence and message count of the distributed scheme, and the effect of connection ratings, and replays every schedule in EPANET and OpenDSS (Secs. X–XI).

The rest of the paper follows that order. Section II reviews related work; Section III sets out the operators, the information available to each, and the sequence of events; Sections IV and V import the water and energy blocks; Section VI states the interface and the integrated benchmark; Sections VII–IX develop the tariff baselines, the contract, and its distributed implementation; Sections X–XII report the case study and discuss it; proofs are in the Appendix.

II. RELATED WORK

Water distribution systems as flexible loads: Optimal operation of water distribution systems has a long history [10], [11]. Pump scheduling against time-varying prices is the classical problem [12]; solution approaches range from dynamic optimization [13] to MILP formulations with piecewise-linear hydraulics [14], Lagrangian decomposition [15], and convex programs for special network classes [16]. Menke *et al.* [3] quantified the demand response a real network can offer. MILPNet [8] generalizes this line of work into a reusable framework that reads an EPANET file and formulates hydraulic elements, control rules, and status checks as MILP constraints; the present paper adopts it as the water operator's private model and uses only its device constraints, since in a day-ahead problem the control rules are the decisions.

Water–power co-optimization: The interface has been studied from the power-system side. Oikonomou *et al.* [4] schedule pumps as a demand-response resource, Oikonomou and Parvania [17] embed a hydraulically detailed model in day-ahead power-system operation, Zamzam *et al.* [18] formulate and distributively solve the joint water–power flow problem, Fooladivanda and Taylor [19] derive convex relaxations for energy-optimal pumping, and Stuhlmacher and Mathieu [20] treat pumping as a recourse resource under uncertainty. Parvania *et al.* [5] formalize the demand-response products a flexible load can offer in a day-ahead market, and Cheng *et al.* [6] schedule a wind–solar–hydro portfolio that bids into the day-ahead market with a reservoir as the flexible element. Most of this literature has a single decision maker: either the utility optimizing against an exogenous price, or a system operator dispatching the water system directly. Settings in which the water asset and the market interface belong to different firms, with different objectives and private information, have received comparatively little attention; [7] is the two-agent LP treatment this paper generalizes.

Distributed energy systems and grid constraints: The energy-hub concept [21], [22] describes conversion and storage among carriers by constant efficiencies, extended with

operational constraints by Evins *et al.* [23]. Morvaj *et al.* [9] embedded the linearized AC power-flow model of Koster and Lemkens [24] in the hub MILP and added a piecewise-linear branch-current limit, so that grid constraints shape the dispatch instead of being checked afterwards. Alternative linear or convex distribution-network models include the linear-programming approximation of AC power flow [25] and the DistFlow model and its relaxations [26], [27]. The formulation of [9] is kept here because it is a MILP with the same piecewise-linear machinery as MILPNet, so both blocks are solved by one solver class and, more to the point, the energy operator's subproblem inherits a well-defined nodal price.

Supply chain coordination with contracts: Decentralized chains of self-interested firms give up system profit, and transfer-payment contracts can restore it. Cachon [28] provides the canonical treatment, building on the double-marginalization insight of Spengler [29] and the channel-coordination results of Jeuland and Shugan [30] and Paster-nack [31]. Revenue-sharing contracts coordinate a newsven-dor channel and divide its profit arbitrarily [32]; Ferrer and Swaminathan [33] analyze two-period product-line models with threshold parameters that partition the strategy space. Applications of this toolkit to the water–energy nexus appear scarce; [7] exploited the energy neutrality of a continuous pump, under which sharing market revenue alone aligns incentives. With on/off pumps that neutrality holds for volume but not for energy, and one contribution of this paper is to show that the contract does not need it.

Decomposition and limited information: Dual decomposition and its price interpretation date to the network utility maximization literature. Kelly *et al.* [34] showed that congestion prices decentralize a network optimum, and Palomar and Chiang [35] systematize primal, dual, and hierarchical decompositions as alternative architectures with their own message-passing patterns; see also [36], [37]. For mixed-integer problems the Lagrangian dual is polyhedral and generally has a gap; bundle methods [38], [39] maximize it far more reliably than subgradient steps, and Lagrangian relaxation with primal recovery is the standard tool for unit commitment [40]. In energy applications decomposition is often motivated by computational scale. Its message-passing structure matters as much here, since it is what lets two organizations reach a joint schedule without sharing forecasts, asset models, or internal state.

III. SYSTEM DESCRIPTION AND THE COORDINATION PROBLEM

A. Agents and Interaction

We consider a site (a municipality, campus, or industrial park) served by a single point of common coupling (PCC) with the distribution grid, behind which two independent operators act (Fig. 1).

The *local water operator* (W) provides an essential service whose electricity use is schedulable because its product can be stored. It runs a water distribution network with reservoirs, pipes, junctions, fixed-speed pumps, valves, and elevated tanks; it must meet an exogenous, inelastic water demand

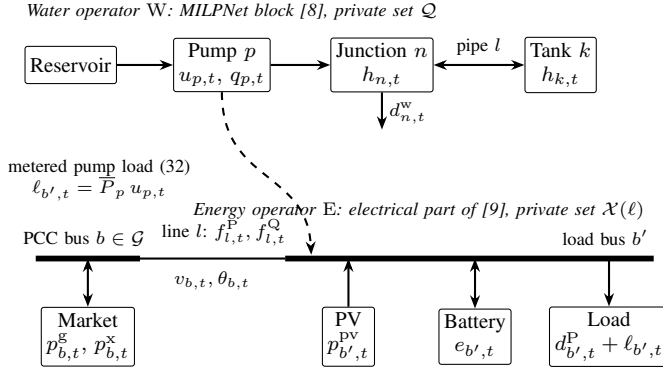


Fig. 1. Two operators, two validated network blocks, one metered interface. The water operator's decisions live in the MILPNet block; the energy operator's in the electrical DES block; the pump load at its supply bus is the only quantity they share.

at every junction and interval, and it treats its hydraulic obligations as hard constraints: minimum service pressures, tank bounds, demand satisfaction, and recovery of the initial storage by the end of the day. Its water revenue and its sourcing costs are fixed by demand, so neither varies with the schedule (Lemma 1). What W controls is *when* each pump runs.

The *local energy operator* (E) owns the site's electrical assets and its market interface. It holds PV generation, a battery, a balanced distribution feeder, and the connection itself, through which it participates in the day-ahead energy market as a price taker [5], [41], buying at c_t^{buy} and selling at $c_t^{\text{sell}} \leq c_t^{\text{buy}}$, and serves the site's other electrical loads. In the prevailing arrangement, which we call the *baseline*, E also acts as W's energy retailer and bills the metered pump load at a regulated tariff.

One relation couples the operators: the pump load at the bus that supplies each pump, which enters E's nodal power balance and is metered by a submeter on the pump feeder. Everything else is internal to one firm or the other. Water demand forecasts, tank levels, pump curves, and hydraulic limits stay with W; load and PV forecasts, battery state and ratings, feeder parameters, and market positions stay with E.

B. Modes of Operation

Three modes organize the analysis. In the *integrated benchmark* a single hypothetical decision maker co-optimizes pumping, storage, dispatch, and market participation subject to both networks; its joint payoff Π^* is the most the site can earn and serves as the yardstick (Sec. VI). In the *uncoordinated baseline* each operator optimizes privately under the retail tariff; the shortfall $\mathcal{G} = \Pi^* - \Pi^0$ is the coordination gain at stake (Sec. VII). In *coordinated operation*, the mode this paper designs, the operators keep their autonomy and their private data but agree on two things: a revenue-sharing contract under which following the jointly optimal schedule is in each operator's own interest (Sec. VIII), and a planning protocol that finds that schedule by exchanging only prices and metered-load schedules (Sec. IX).

C. Sequence of Events

Following the contracting convention of [28], the interaction unfolds in four stages. (1) *Contracting*: the operators agree once on the terms of trade, a share φ and a fixed fee F . (2) *Planning*: on the day before delivery they coordinate schedules by exchanging messages; E then submits its day-ahead position before gate closure. (3) *Delivery*: schedules are executed; the meter at the PCC and the submeter on the pump feeder record actual flows. (4) *Settlement*: the market settles the position and the contract transfer is paid based on the settlement statement and the meter readings. Two kinds of information should be distinguished: planning-stage information stays private; settlement-stage quantities are metered and verifiable, so a contract can be written on them. The planning protocol uses the first kind; the contract uses only the second.

D. Assumptions

Assumption 1 (Deterministic price taker). *Prices, water demand, PV availability, and the non-pump electrical loads are known point forecasts for the horizon. The operators are small relative to the market, so their positions do not move the price, and at the planning stage energy scheduled for delivery is treated as delivered exactly. Imbalance settlement of delivery deviations is discussed in Remark 2.*

Assumption 2 (Fixed-speed pumps). *Pumps are fixed-speed units with binary status; a running pump draws a calibrated constant electrical power $\bar{P}_p$ (Sec. VI). Its flow and head follow its characteristic curve and the network hydraulics.*

Assumption 3 (Curtailed PV, exclusive modes). *PV can be curtailed within its availability, the battery cannot charge and discharge in the same interval, and the connection cannot import and export in the same interval, as in [9]. Surplus that can be neither consumed, stored, nor exported is spilled at no cost.*

Assumption 4 (Adequacy). *The feeder and the connection can serve the site's load together with any feasible pump schedule, so that the baseline is well defined and comparisons against it are meaningful. The case study tests the boundary of this assumption by tightening the connection ratings.*

The coordination question is then concrete: the jointly optimal schedule maximizes the joint payoff, but the schedule is chosen by two self-interested operators under the information structure above. What terms of trade make following the jointly optimal schedule worth each operator's while, how is the surplus divided, and how is the schedule computed without pooling private data?

IV. THE WATER OPERATOR'S BLOCK

The water operator's private constraint set $\mathcal{Q}$ is the MILPNet formulation of Thomas and Sela [8]. A directed graph contains junctions $\mathcal{J}$, tanks $\mathcal{K}$, reservoirs $\mathcal{R}$, and links $\mathcal{L}$ consisting of pipes $\mathcal{L}^p$, pumps $\mathcal{P}$, and valves $\mathcal{V}$. Link orientation fixes the sign of flow; pipes and gate valves may carry flow in either direction, whereas pumps do not allow backflow. Reservoir

TABLE I
CONSTRAINTS OF THE TWO BLOCKS AND THEIR ORIGIN

Constraint	Source	Treatment in this paper
<i>Water operator (Q)</i>		
Junction mass balance (1)	T&S (1)	As published
Pipe energy conservation (2)	T&S (2)	As published
Hazen–Williams loss (3)	T&S (3)	PWL over a per-pipe signed domain
Tank dynamics (7)	T&S (4)	Plus bounds and terminal reserve
Pump curve and status (10)–(12)	T&S (5)–(7)	Status 1 = on; closure slack eliminated; admissible flow range
Gate valve (13)	T&S (8), (11)	As published
PRV states (14)	T&S (9), (12)	As published; not exercised
Tank-link checks, control rules	T&S (13)–(15)	Omitted: statuses are decisions
Pressure floor (4)	—	Added service constraint
<i>Energy operator (X)</i>		
Electricity balance (16)	M (1), (14)	Nodal form; metered pump load added
Heat balance	M (2)	Omitted: no thermal coupling
Storage continuity (18)	M (3)	Efficiencies applied once
Capacity and exclusivity (21)–(23)	M, Sec. 2.2	As described; terminal closure added
Linearized AC flow (24)–(25)	M (10), (11)	$U_0 = 1$ p.u.
Reactive balance (17)	M (15)	Grid supplies reactive power
Branch current (26)–(29)	M (25)–(31)	SOS2 interpolation; no objective weight
Settlement (30)	M (4)	Electricity streams only; sale price added
<i>Interface</i>		
Metered pump load (32), calibration (33)	—	This paper; relaxed in Sec. IX

heads follow their prescribed time series, $h_{r,t} = H_{r,t}$ for $r \in \mathcal{R}$. Table I maps every constraint of both blocks to its source.

A. Conservation Laws

Mass balance at every junction and interval requires that the net inflow equal the demand [8, Eq. (1)]:

$$\sum_{l \in \mathcal{L}} A_{nl}^w q_{l,t} = d_{n,t}^w, \quad n \in \mathcal{J}, \quad t \in \mathcal{T}. \quad (1)$$

Demand is the sum over all demand categories evaluated with the network's time patterns; consumption is demand driven, so pressure-dependent demand and leakage are not represented. Energy conservation along a pipe $l = (i, j)$ relates the head difference to the head loss [8, Eq. (2)],

$$h_{i,t} - h_{j,t} - \Delta h_{l,t} = 0, \quad l \in \mathcal{L}^p, \quad t \in \mathcal{T}, \quad (2)$$

and the head loss follows the Hazen–Williams relation [8, Eq. (3)],

$$\Delta h_{l,t} = \text{sgn}(q_{l,t}) K_l |q_{l,t}|^{1.852}, \quad K_l = \frac{10.67 L_l}{C_l^{1.852} D_l^{4.8704}}, \quad (3)$$

with pipe length L_l (m), diameter D_l (m), and roughness C_l . Equation (3) is nonlinear and nonconvex because of the bidirectional flow, and MILPNet replaces it by a piecewise-linear (PWL) interpolation of the signed curve over a bounded flow domain. A service constraint that MILPNet leaves to the user is added here as a minimum pressure head at every junction:

$$h_{n,t} - z_n \geq \underline{p}_n, \quad n \in \mathcal{J}, \quad t \in \mathcal{T}. \quad (4)$$

B. Piecewise-Linear Representation

All univariate nonlinearities in both blocks use the same construction. Given breakpoints (x_s, y_s) , $s = 0, \dots, S$, with strictly increasing x_s , the relation $y = \text{PWL}(x)$ is represented by nonnegative weights λ_s and segment binaries δ_s , $s = 0, \dots, S-1$ [42]–[44]:

$$x = \sum_{s=0}^S \lambda_s x_s, \quad y = \sum_{s=0}^S \lambda_s y_s, \quad \sum_{s=0}^S \lambda_s = \sum_{s=0}^{S-1} \delta_s = \zeta, \quad (5)$$

$$\lambda_s \leq \delta_{s-1} + \delta_s, \quad \delta_s \in \{0, 1\}, \quad (6)$$

where undefined end binaries are zero and ζ is an activation term. For pipes and lines $\zeta = 1$; for a pump $\zeta = u_{p,t}$, so that both the flow and the head gain vanish when the pump is off. Adjacent breakpoints only can carry positive weight, which is the special-ordered-set (SOS2) condition written with explicit binaries; the formulation is therefore independent of solver-specific SOS or indicator support. For the Hazen–Williams curve, the breakpoints span a signed domain $[-\bar{q}_l, \bar{q}_l]$ with zero as a breakpoint; following MILPNet, the domain is set from an extended-period simulation of the network so that the segments cover the actual operating range. The interpolation is an equality surrogate of the nonlinear relation, not a relaxation with a guaranteed hydraulic error bound; the case study reports the resulting discrepancies through replay.

C. Tanks and Inventory

For a cylindrical tank k the state is its total head, which evolves with the net inflow over each interval [8, Eq. (4)]:

$$h_{k,t+1} = h_{k,t} + \frac{\Delta t_s}{A_k} \sum_{l \in \mathcal{L}} A_{kl}^w q_{l,t}, \quad k \in \mathcal{K}, \quad t \in \mathcal{T}, \quad (7)$$

$$h_{k,0} = h_k^0, \quad \underline{h}_k \leq h_{k,t} \leq \bar{h}_k, \quad k \in \mathcal{K}, \quad t \in \bar{\mathcal{T}}, \quad (8)$$

$$h_{k,T} \geq h_k^0, \quad k \in \mathcal{K}. \quad (9)$$

The inventory is defined on all $T+1$ boundaries, so the withdrawal or filling decided in the last interval changes $h_{k,T}$ and cannot disappear from the accounting, and the terminal condition (9) preserves the initial reserve, which prevents apparent savings obtained by draining the tank; all compared schedules use the same terminal requirement. MILPNet's tank-link status checks [8, Eq. (13)] are replaced by the hard bounds (8): a schedule that respects the bounds at every boundary does not trigger those checks in the discretized model, and whether it does so within an interval is a question for the replay of Section XI.

D. Pumps

A fixed-speed pump $p = (i, j)$ adds head according to a head–flow curve when it operates [8, Eq. (5)]. EPANET's power-law form $g_p(q) = a_p - b_p q^{\nu_p}$ is used so that single-point and three-point curves are covered by the same expression, and g_p is interpolated with the construction of Section IV-B:

$$\gamma_{p,t} = \text{PWL}_p(q_{p,t}) \text{ with } \zeta = u_{p,t}, \quad p \in \mathcal{P}, \quad t \in \mathcal{T}. \quad (10)$$

The interpolation domain is the hydraulically admissible flow range $[0, \bar{q}_p]$: a running pump must lift water at least from the

highest head its suction node can take to the lowest head its discharge node may have (pressure floor or tank bottom), so $\bar{q}_p$ is the curve flow at that minimum gain rather than the zero-head flow. Flows beyond it are never feasible, and excluding them tightens the LP relaxation without changing the feasible set. MILPNet models the two states of a pump with a closure slack on the energy balance and a big- M pair [8, Eqs. (6)–(7)]. Eliminating the slack gives the equivalent form

$$0 \leq q_{p,t} \leq \bar{q}_p u_{p,t}, \quad (11)$$

$$-M_p(1 - u_{p,t}) \leq h_{j,t} - h_{i,t} - \gamma_{p,t} \leq M_p(1 - u_{p,t}), \quad (12)$$

for $p \in \mathcal{P}$, $t \in \mathcal{T}$. When $u_{p,t} = 1$ the downstream head equals the upstream head plus the curve gain; when $u_{p,t} = 0$ the flow and the gain are zero and the two end heads are decoupled. The status convention is the complement of MILPNet's, where $y_p^t = 0$ denotes an open pump; the convention used here makes the interface of Section VI a product of the status with a positive power. Following MILPNet, M_p is set just above the largest admissible head: since head is added only by pumps and decreases along the direction of flow, no node can sit above the highest source head plus the sum of the pump shutoff heads, and that ceiling bounds every head variable and every M_p .

E. Valves

Gate valves and pressure-reducing valves (PRVs) are retained from MILPNet for networks that contain them; the case study network has none, so these constraints are stated compactly. A gate valve $g = (i, j)$ with open status $y_{g,t}$ behaves like a pipe when open and disconnects its ends when closed [8, Eqs. (8), (11)]:

$$\begin{aligned} -\bar{q}_g y_{g,t} &\leq q_{g,t} \leq \bar{q}_g y_{g,t}, \\ -M_g(1 - y_{g,t}) &\leq h_{i,t} - h_{j,t} - \Delta h_{g,t} \leq M_g(1 - y_{g,t}), \end{aligned} \quad (13)$$

with $\Delta h_{g,t}$ the PWL Hazen–Williams loss of the valve. A PRV $v = (i, j)$ with setting H_v^{set} is in exactly one of three states, active ($\alpha_{v,t}$), open ($\omega_{v,t}$), or closed ($\kappa_{v,t}$) [8, Eqs. (9), (12)]:

$$\begin{aligned} \alpha_{v,t} + \omega_{v,t} + \kappa_{v,t} &= 1, \quad 0 \leq q_{v,t} \leq \bar{q}_v(1 - \kappa_{v,t}), \\ |h_{j,t} - H_v^{\text{set}}| &\leq M_v(1 - \alpha_{v,t}), \quad h_{i,t} \geq H_v^{\text{set}} - M_v(1 - \alpha_{v,t}), \\ |h_{i,t} - h_{j,t}| &\leq M_v(1 - \omega_{v,t}), \quad h_{i,t} \leq H_v^{\text{set}} + M_v(1 - \omega_{v,t}), \\ h_{i,t} - h_{j,t} + \epsilon \kappa_{v,t} &\leq M_v(1 - \kappa_{v,t}), \end{aligned} \quad (14)$$

where each absolute value denotes the corresponding pair of linear inequalities and $\epsilon > 0$ replaces the strict inequality of the closed state, as in MILPNet.

F. Control Rules

MILPNet also formulates event-based and time-based control rules [8, Eqs. (14)–(15)], which make the MILP reproduce the rule-based behavior of an extended-period simulation. In the day-ahead problem the pump statuses are the decisions, so those rules are omitted; the rule-based schedule of the input file is used only for calibration (Section VI). MILPNet's pump-scheduling objective [8, Eq. (16)] contains a fixed energy cost

per operating hour and a switching penalty. The energy cost is what the arrangements of Sections VII–IX replace; the switching penalty is not included in the present implementation, but it is linear in $u_{p,t}$ and can be added without changing the problem class.

The water operator's private set is

$$\mathcal{Q} = \{(q, h, u, \gamma, \dots) : (1)–(4), (7)–(14), (5)–(6)\}, \quad (15)$$

a nonconvex set because of the binaries; nothing in the contract argument of Section VIII uses its convexity.

V. THE ENERGY OPERATOR'S BLOCK

The energy operator's private set is the DES model of Morvaj *et al.* [9], whose energy-hub layer assigns conversion and storage technologies to meet electricity and heat demand while a linearized AC power-flow layer represents the distribution grid. Only the electrical part is imported. Each electrical bus $b \in \mathcal{B}$ hosts a hub with, at most, a PV generator, a battery, non-pump loads, and, if $b \in \mathcal{G}$, a grid connection. Conversion technologies with heat output and the thermal storage are absent, so the conversion-efficiency matrix of the hub reduces to the identity for PV and the heat balance [9, Eq. (2)] is dropped. The asset capacities are given, not designed.

A. Electricity Balance

The hub electricity balance [9, Eq. (1)] and the nodal active-power balance of the grid layer [9, Eq. (14)] are combined into one balance per bus, with the metered pump load $\ell_{b,t}$ of Section VI added to the load:

$$\begin{aligned} p_{b,t}^g - p_{b,t}^x + p_{b,t}^{\text{pv}} + p_{b,t}^{\text{dis}} - p_{b,t}^{\text{ch}} + \sum_{l \in \mathcal{E}} A_{bl}^e f_{l,t}^{\text{P}} \\ = d_{b,t}^{\text{P}} + \ell_{b,t}, \quad b \in \mathcal{B}, \quad t \in \mathcal{T}. \end{aligned} \quad (16)$$

The reactive balance [9, Eq. (15)] is

$$r_{b,t}^g + \sum_{l \in \mathcal{E}} A_{bl}^e f_{l,t}^{\text{Q}} = d_{b,t}^{\text{Q}}, \quad b \in \mathcal{B}, \quad t \in \mathcal{T}, \quad (17)$$

where PV, battery, and pump exchanges are at unity power factor and reactive load is supplied from the grid connection; $r_{b,t}^g = 0$ for $b \notin \mathcal{G}$.

B. Storage, Generation, and Grid Exchange

Storage continuity [9, Eq. (3)] is written with charging and discharging measured at the AC bus, so that the efficiencies appear once, in the inventory equation, and not again in the bus balance:

$$e_{b,t+1} = e_{b,t} + \Delta t \left(\eta_b^{\text{ch}} p_{b,t}^{\text{ch}} - \frac{p_{b,t}^{\text{dis}}}{\eta_b^{\text{dis}}} \right), \quad t \in \mathcal{T}, \quad (18)$$

$$e_{b,0} = e_b^0, \quad 0 \leq e_{b,t} \leq \bar{e}_b, \quad t \in \bar{\mathcal{T}}, \quad (19)$$

$$e_{b,T} \geq e_b^0. \quad (20)$$

As for tanks, the inventory is tracked on all $T + 1$ boundaries and the terminal energy must be replenished. PV can be curtailed within its availability,

$$0 \leq p_{b,t}^{\text{pv}} \leq \bar{p}_{b,t}^{\text{pv}}, \quad (21)$$

and, as required in [9], a storage cannot charge and discharge, nor can a bus import and export, in the same interval:

$$0 \leq p_{b,t}^{\text{ch}} \leq \bar{p}_b^{\text{ch}} y_{b,t}^{\text{ch}}, \quad 0 \leq p_{b,t}^{\text{dis}} \leq \bar{p}_b^{\text{dis}} y_{b,t}^{\text{dis}}, \quad y_{b,t}^{\text{ch}} + y_{b,t}^{\text{dis}} \leq 1, \quad (22)$$

$$0 \leq p_{b,t}^{\text{g}} \leq \bar{p}_b^{\text{g}} y_{b,t}^{\text{g}}, \quad 0 \leq p_{b,t}^{\text{x}} \leq \bar{p}_b^{\text{x}} y_{b,t}^{\text{x}}, \quad y_{b,t}^{\text{g}} + y_{b,t}^{\text{x}} \leq 1, \quad (23)$$

for $b \in \mathcal{B}$, $t \in \mathcal{T}$, with $p_{b,t}^{\text{g}} = p_{b,t}^{\text{x}} = 0$ for $b \notin \mathcal{G}$ and zero storage and PV bounds at buses without those assets. The ratings $\bar{p}^{\text{g}}, \bar{p}^{\text{x}}$ of the connection are the “interconnection limits” of the two-agent model. Battery degradation is not modeled.

C. Linearized AC Power Flow

Each line $l = (i, j)$ has series impedance $R_l + jX_l$ and series admittance $G_l + jB_l$, with $G_l = R_l/(R_l^2 + X_l^2)$ and $B_l = -X_l/(R_l^2 + X_l^2)$. Morvaj *et al.* adopt the linearization of [24], a first-order expansion of the AC power-flow equations about a flat voltage profile U_0 with small angle differences [9, Eqs. (10)–(11)]. With $U_0 = 1$ p.u. and the power base S_B ,

$$f_{l,t}^{\text{P}} = S_B [G_l(v_{i,t} - v_{j,t}) - B_l(\theta_{i,t} - \theta_{j,t})], \quad (24)$$

$$f_{l,t}^{\text{Q}} = S_B [-B_l(v_{i,t} - v_{j,t}) - G_l(\theta_{i,t} - \theta_{j,t})], \quad (25)$$

for $l \in \mathcal{E}$, $t \in \mathcal{T}$. Voltage magnitudes obey $\underline{v} \leq v_{b,t} \leq \bar{v}$, and the reference bus has $v = 1$ and $\theta = 0$. The model is balanced, lossless, and topology fixed; a solution of (24)–(25) is therefore not a certificate of AC feasibility, which is why the optimized injections are replayed in a nonlinear power-flow solver in Section XI.

D. Branch-Current Limits

Morvaj *et al.* linearize the branch current by the same expansion, obtaining real and imaginary parts that are linear in the voltage and angle differences [9, Eq. (25)]. At the expansion point these components coincide with the per-unit line flows of (24)–(25):

$$i_{l,t}^{\text{re}} = G_l(v_{i,t} - v_{j,t}) - B_l(\theta_{i,t} - \theta_{j,t}) = f_{l,t}^{\text{P}}/S_B, \quad (26)$$

$$i_{l,t}^{\text{im}} = B_l(v_{i,t} - v_{j,t}) + G_l(\theta_{i,t} - \theta_{j,t}) = -f_{l,t}^{\text{Q}}/S_B. \quad (27)$$

Following [9, Eqs. (26)–(31)], nonnegative auxiliary variables bound the absolute values,

$$\phi_{l,t} \geq \pm i_{l,t}^{\text{re}}, \quad \chi_{l,t} \geq \pm i_{l,t}^{\text{im}}, \quad (28)$$

their squares are interpolated on $[0, \bar{v}_l]$ with the construction of Section IV-B, and the interpolated squares are limited:

$$\hat{\phi}_{l,t} = \text{PWL}(\phi_{l,t}^2), \quad \hat{\chi}_{l,t} = \text{PWL}(\chi_{l,t}^2), \quad \hat{\phi}_{l,t} + \hat{\chi}_{l,t} \leq \bar{v}_l^2. \quad (29)$$

The adjacency binaries of (6) are kept so that the same interpolation code serves both blocks, and no objective perturbation is needed: increasing $\phi_{l,t}$ or $\chi_{l,t}$ beyond the absolute value can only tighten (29). Because each secant lies above x^2 , the constraint is conservative for the magnitude of the linearized current.

E. Settlement

The DES objective of [9, Eq. (4)] is a weighted sum of input energy streams. With natural gas absent and the weights equal to the day-ahead prices, it becomes the energy operator’s day-ahead settlement, revenue from exports net of the cost of imports:

$$R = \Delta t \sum_{t \in \mathcal{T}} \sum_{b \in \mathcal{G}} (c_t^{\text{sell}} p_{b,t}^{\text{x}} - c_t^{\text{buy}} p_{b,t}^{\text{g}}). \quad (30)$$

For a given metered pump load $\ell = (\ell_{b,t})$ the energy operator’s private set is

$$\mathcal{X}(\ell) = \{(p, e, f, v, \theta, i, \dots) : (16)–(29), (5)–(6)\}, \quad (31)$$

and $R_E(\ell) = \max\{R : \xi \in \mathcal{X}(\ell)\}$ is its best settlement when it serves that load.

VI. INTERFACE AND INTEGRATED BENCHMARK

A. Metered Pump Load

The two blocks share one quantity. Each running pump draws a constant electrical power $\bar{P}_p$ at the bus that supplies it, and the metered load at bus b is

$$\ell_{b,t} = \sum_{p \in \mathcal{P}_b} \bar{P}_p u_{p,t}, \quad b \in \mathcal{B}, \quad t \in \mathcal{T}, \quad (32)$$

which enters the bus balance (16). The coefficient is calibrated from an extended-period EPANET simulation of the input network under its original controls, the same simulation that MILPNet uses to bound the PWL domains. With $q_{p,\tau}^{\text{sim}}$ and $\Delta h_{p,\tau}^{\text{sim}}$ the simulated flow and head gain of pump p , and $\mathcal{O}_p$ the set of simulation steps in which it runs,

$$\bar{P}_p = \frac{\rho g}{\eta_p} \cdot \frac{1}{|\mathcal{O}_p|} \sum_{\tau \in \mathcal{O}_p} q_{p,\tau}^{\text{sim}} \Delta h_{p,\tau}^{\text{sim}}, \quad (33)$$

with $\rho g = 9.81 \text{ kW}\cdot\text{s}/\text{m}^4$ for water, rounded to 10 W. Equation (32) is linear in the existing pump status, so the coupled problem introduces no new nonlinearity or integrality. It is also the principal coupling approximation: the modeled electrical draw depends on whether the pump runs, not on the optimized flow and head. Two features of (32) matter for what follows. It is *metered*: a submeter on the pump feeder records it, so it is a settlement-grade quantity. And it is the *only* constraint that involves both operators’ variables.

B. Integrated Benchmark

The integrated problem co-optimizes both operators’ decisions,

$$\Pi^* = \max \{R : (q, h, u, \dots) \in \mathcal{Q}, \quad \xi \in \mathcal{X}(\ell), (32)\}, \quad (34)$$

a MILP whose binaries are the pump statuses, valve states, storage and grid modes, and the PWL segment selectors. Three properties follow from the construction. First, tanks and batteries have the same accounting role: each can move consumption across time but neither can supply net energy or water through an unrecorded terminal withdrawal, because of (9) and (20). Second, with exact solves, fixing the pump statuses to any feasible schedule cannot increase R , so $\Pi^* - R_E(\ell(u)) \geq 0$ for

every $u \in \mathcal{Q}$: the value of coordination against any schedule is a well-defined nonnegative number. Third, the site's cost is dominated by the fixed loads, so the interesting quantities are differences between arrangements, not levels.

C. Water-Side Accounting

In the two-agent model of [7] a continuous pump with constant specific energy makes every feasible schedule use the same energy, which is what allows revenue sharing alone to coordinate. Here the volume identity survives but the energy identity does not.

Lemma 1 (Volume neutrality). *Every feasible schedule delivers the demanded volume plus the net change of tank storage: summing (1) over $\mathcal{J}$ and (7) over $\mathcal{K}$ and $\mathcal{T}$,*

$$\begin{aligned} \Delta t_s \sum_{t \in \mathcal{T}} \sum_{r \in \mathcal{R}} \sum_{l \in \mathcal{L}} (-A_{rl}^w) q_{l,t} \\ = \Delta t_s \sum_{t \in \mathcal{T}} \sum_{n \in \mathcal{J}} d_{n,t}^w + \sum_{k \in \mathcal{K}} A_k (h_{k,T} - h_k^0), \end{aligned}$$

where the left side is the volume drawn from the sources. Under (9) the storage term is nonnegative, and it is at its smallest feasible value at every schedule that no positive electricity price improves. The water operator's volume-proportional revenues and costs (sales, sourcing, treatment) are therefore invariant across candidate schedules up to the terminal storage granularity of the on/off pump, and are omitted from its payoff. Its pumping energy $\Delta t \sum_p \bar{P}_p \sum_t u_{p,t}$ is not invariant: the flow a pump delivers in an hour depends on the head it pumps against, so schedules can differ in their number of pump-hours.

The consequence is that the water operator's schedule-dependent payoff is, under any arrangement, exactly its energy-related cashflow: the bill it pays under a tariff, or the transfer it receives under a contract. The energy operator's schedule-dependent payoff is its settlement (30) plus whatever it collects from W. The integrated payoff is $\Pi = R$.

VII. BASELINE AND WHY TARIFFS FAIL

In the baseline W buys its pumping energy from E at a retail tariff τ_t and solves

$$\min_{(q,h,u,\dots) \in \mathcal{Q}} \Delta t \sum_{t \in \mathcal{T}} \sum_{p \in \mathcal{P}} \tau_t \bar{P}_p u_{p,t}, \quad (35)$$

while E, taking the metered load $\ell^0 = \ell(u^0)$ as fixed, solves its market problem $R_E(\ell^0)$. Baseline payoffs are $\Pi_W^0 = -\text{bill}^0$ and $\Pi_E^0 = R_E(\ell^0) + \text{bill}^0$, where $\text{bill}^0 = \Delta t \sum_t \sum_p \tau_t \bar{P}_p u_{p,t}^0$; the bill cancels in the joint value $\Pi^0 = R_E(\ell^0)$, and the coordination gain is $\mathcal{G} = \Pi^* - \Pi^0 \geq 0$.

Under a flat tariff $\tau_t \equiv \bar{\tau}$ the objective of (35) is $\bar{\tau} \Delta t \sum_p \bar{P}_p \sum_t u_{p,t}$, which depends on the schedule only through its pump-hours. The tariff therefore selects the *energy-minimal* schedules, those with the least pump-hours that serve demand and replenish the tank, and leaves their timing to operating practice. The discrete analogue of the “float” operation of [7], pumping as demand arrives and keeping the tank near

its setpoint, is *level holding*: among the energy-minimal schedules, the one that minimizes the summed absolute deviation of the tank level from its initial value. It is sound practice, and it is what a water operator with no timing signal does. The baseline of the case study is computed in two stages, the minimal pump-hour count first and the level-holding schedule among schedules with that count second, so that the tie-break is exact rather than left to the solver's tolerance.

Proposition 1 (Tariff arrangements do not coordinate). *Let $\mathcal{G} > 0$ and let π^* be the internal price of Proposition 2.*

- (i) *Under a flat tariff, every energy-minimal schedule is a best response of W: the tariff cannot single out the timing the market rewards, and the schedule operating practice adopts captures $\mathcal{G}$ only by coincidence. If the jointly optimal schedule uses more pump-hours than the minimum, no flat tariff makes it a best response.*
- (ii) *Under the pass-through tariff $\tau_t = c_t^{\text{buy}} + \eta$, W's best responses minimize $\sum_t c_t^{\text{buy}} \ell_t$. A jointly optimal schedule is among them only if $\pi_t^* = c_t^{\text{buy}}$ in every hour it pumps, which fails whenever it pumps in an export, idle, or congested hour; and among its best responses W is indifferent between hours of equal price.*
- (iii) *Under the internal-price tariff $\tau_t = \pi_t^* + \eta$, the jointly optimal schedule is a best response of W whenever the Lagrangian relaxation of (32) has no duality gap, but ties among equally priced hours remain, and the division of $\mathcal{G}$ is dictated by (π^*, η) through the bill; participation of both operators is not guaranteed.*

Part (i) is the operational analogue of double marginalization [28], [29]: a price that does not vary where the site's does gives the downstream firm no reason to act in the chain's interest. Part (ii) says that even the market price is the wrong signal in the hours that matter most for a site with local generation: when the site exports or curtails, energy at the site is worth the sale price or nothing, and a water operator charged the purchase price will not pump into the surplus. Part (iii) says that the right hourly price still cannot separate schedules of equal price, of which the on/off pump produces many, and fixes the split of the gain as a side effect of its level. What is missing is an instrument that divides the gain without disturbing the hourly signal and that makes the operators agree on one schedule. That is the role of revenue sharing and of the planning protocol.

VIII. THE REVENUE-SHARING CONTRACT

A. Contract and Incentives

Computing the jointly optimal schedule is an optimization problem; making two independent firms follow it is not. Once the tariff is replaced by contract terms, each operator runs whatever schedule maximizes its own payoff under those terms. The design question is therefore one of incentives: when one operator follows the jointly optimal schedule, the other can do no better than to follow it too. That property, no profitable unilateral deviation, is a Nash equilibrium and the standard test of a coordinating contract [28]. We propose the following terms of trade in place of the retail tariff.

Definition 1 (Day-ahead revenue-sharing contract). *Under contract $\mathcal{C}(\varphi, F)$ with share $\varphi \in [0, 1]$ and fixed fee $F \in \mathbb{R}$, the transfer from E to W at settlement is*

$$T(\varphi, F) = \varphi R - F, \quad (36)$$

where R is the settlement (30) of the metered day-ahead position. Payoffs are $\Pi_W = T$ and $\Pi_E = R - T$; both operators' water-side revenues and costs are fixed by demand (Lemma 1) and are omitted.

The two terms play different roles. The share φ is what changes behavior: it gives W an interest in the settlement its pumping schedule helps create. The fee F changes no one's behavior, but it is not optional: it is what makes both operators willing to sign, and it is the instrument that divides the gain. The contract requires no cost audit and no exchange of planning data. R is verifiable from the market settlement statement and the pump load from the submeter.

Theorem 1 (Revenue sharing coordinates). *Let Assumptions 1–4 hold and let (u^*, ξ^*) be an integrated optimum of (34). Consider the game in which W chooses $u \in \mathcal{Q}$ and E chooses $\xi \in \mathcal{X}(\ell(u))$, a unilateral deviation being available to an operator only if the resulting operating point remains feasible with the other's decisions fixed. Then, for every $\varphi \in [0, 1]$ and every $F \in \mathbb{R}$:*

- (i) (u^*, ξ^*) is a Nash equilibrium of $\mathcal{C}(\varphi, F)$: the contract coordinates in the sense of [28];
- (ii) its payoffs are $\Pi_W = \varphi \Pi^* - F$ and $\Pi_E = (1 - \varphi) \Pi^* + F$, and as (φ, F) range over $[0, 1] \times \mathbb{R}$ the pair attains every division of Π^* .

The proof (Appendix) compares payoffs at feasible points and nowhere uses convexity, so the on/off pumps, the PWL hydraulics, and the exclusivity binaries of both blocks leave it unchanged. Nor does it use energy neutrality: the newsvendor obstacle to pure revenue sharing, a downstream firm with private unit costs that the contract distorts [32], is absent because the water operator's only schedule-dependent cashflow is the transfer itself; its volume-proportional costs cannot be distorted (Lemma 1), and its pumping energy is already inside R . As with coordinating contracts generally, the equilibrium need not be unique when several equally good schedules exist; the planning protocol of Section IX picks the one the operators agree to run, and Theorem 1 then guarantees that once adopted it is self-enforcing.

B. Dividing the Gain

For a site with more generation than load the settlement is positive and the natural anchor for the fee is the baseline bill, which yields a window of shares $[0, \mathcal{G}/R^*]$ in which both operators gain [7]. The site of this paper, like most municipal sites, is a net importer: $R^* < 0$, and with $F = \text{bill}^0$ every positive share would hand W a share of a net cost. The fee absorbs the difference.

Corollary 1 (Fee anchoring and the Pareto window). *Let $R^0 = \Pi^0$ and $R^* = \Pi^*$. For every share $\varphi \in [0, 1]$ the contract Pareto-dominates the baseline exactly for fees*

$$F \in [\text{bill}^0 + R^0 - (1 - \varphi)R^*, \text{bill}^0 + \varphi R^*], \quad (37)$$

an interval of width $\mathcal{G}$. The anchored fee $F(\varphi) = \text{bill}^0 + \varphi R^0$ lies in it and gives $\Pi_W = -\text{bill}^0 + \varphi \mathcal{G}$ and $\Pi_E = \Pi_E^0 + (1 - \varphi)\mathcal{G}$: the water operator's payoff rises by $\varphi \mathcal{G}$ and the energy operator's by $(1 - \varphi)\mathcal{G}$, whatever the sign of the settlement. The equal split, the Nash bargaining outcome when each operator's fallback is its baseline payoff [45], is $\varphi = 1/2$ with the anchored fee.

For a net importer the contract admits a plain reading. Since $R = -C$ with C the site's day-ahead bill, (36) says that W pays E the amount $\varphi C + F$: a share of the site's bill plus a fee. With the anchored fee, W pays $\text{bill}^0 - \varphi(C^0 - C^*)$, its old bill less its share of the savings. The share is what makes W care about the timing of the site's bill; the rebate is what returns the baseline difference. Neither term requires E to open its books, and neither requires W to disclose a demand forecast.

Remark 1 (Discrete commitments). *Theorem 1 compares payoffs at feasible points, so the integer pump commitments, valve states, and PWL selectors affect only the computation of the integrated optimum, not the incentive argument. What the discreteness does change is the distributed computation: the Lagrangian dual of (32) acquires a duality gap (Proposition 3), and the tariff of Proposition 1(iii) no longer selects a unique schedule.*

Remark 2 (Imbalance settlement). *Under Assumption 1 delivered energy equals the day-ahead position. Under a two-settlement market design, delivered energy earns the day-ahead price while deviations from the committed position are settled at an imbalance price [6]; the transfer (36) then shares the net settled revenue, and since it is a fixed share of whatever revenue materializes plus a constant, the incentive argument of Theorem 1 applies to each realization separately. The water operator becomes a partial equity holder in the market position; instruments to reshape the risk exist within the same family (a larger fee with a smaller share, floors or collars on the transfer), and designing that split with risk preferences requires a stochastic model, left to future work.*

IX. DISTRIBUTED IMPLEMENTATION WITH LIMITED INFORMATION

Theorem 1 makes the jointly optimal schedule self-enforcing, but someone must still *find* it without pooling the operators' private data. Problem (34) consists of two independently constrained blocks, $\mathcal{Q}$ and $\mathcal{X}$, tied by the single interface (32). Relaxing that constraint with a price splits the problem into two subproblems, each solved by its own operator from its own data, and the price is adjusted until the two schedules agree at the meter. Solving (34) centrally would instead require one party to hold both operators' constraint sets and forecasts, which is what the organizational separation rules out.

A. Price Decomposition

Relax (32) with multipliers $\pi = (\pi_{b,t})_{b \in \mathcal{B}_P, t \in \mathcal{T}}$, the *internal price* at every bus that supplies a pump, and let $w_{b,t} \in [0, \sum_{p \in \mathcal{P}_b} \bar{P}_p]$ denote the pump load E accepts to serve at that price. The Lagrangian separates into a water subproblem that touches only W's private data,

$$\ell(\pi) \in \arg \min_{(q,h,u,\dots) \in \mathcal{Q}} \Delta t \sum_{t \in \mathcal{T}} \sum_{p \in \mathcal{P}} \pi_{b(p),t} \bar{P}_p u_{p,t}, \quad (\text{P}_W)$$

where $b(p)$ is the bus supplying pump p , and an energy subproblem that touches only E's,

$$w(\pi) \in \arg \max_{\xi \in \mathcal{X}(w), w} R(\xi) + \Delta t \sum_{t \in \mathcal{T}} \sum_{b \in \mathcal{B}_P} \pi_{b,t} w_{b,t}. \quad (\text{P}_E)$$

Subproblem (P_W) is W's ordinary price-based pump-scheduling problem, the same MILP it would solve under any hourly tariff: it reuses (35) with τ replaced by the internal price. Subproblem (P_E) is E's dispatch problem with a price-responsive load that it is paid π to serve. The dual function

$$D(\pi) = \Delta t \sum_{t,p} \pi_{b(p),t} \bar{P}_p u_{p,t}(\pi) - R(\xi(\pi)) - \Delta t \sum_{t,b} \pi_{b,t} w_{b,t}(\pi) \quad (38)$$

is concave and piecewise linear, bounds the integrated cost from below, $D(\pi) \leq -\Pi^*$, and has the metered imbalance $g_{b,t}(\pi) = \ell_{b,t}(\pi) - w_{b,t}(\pi)$ as a supergradient. Everything E needs to run the master problem is the load schedule W sends: W's subproblem value is $\Delta t \sum \pi \ell$, computable by E from the metered schedule and the price it posted.

B. Master Problem and Primal Recovery

Algorithm 1 summarizes the scheme with E, the owner of the PCC, acting as the master. The first message deserves attention. In the LP model the market price is a natural start, since it is already the internal price in every uncongested hour [7]; here it is a poor one, because at $\pi = c^{\text{buy}}$ the energy operator is indifferent about serving the pump in every import hour and its subproblem returns an arbitrary accepted load. A better first message is available to E from its own data: the marginal cost of serving the metered baseline schedule, i.e., the multipliers of its nodal balances when it dispatches against that schedule. By Proposition 2 this already equals the internal price in every hour in which coordination does not change the state of the connection, so the scheme starts close to the optimum and the master only has to correct the hours in which it does. Two update rules are implemented. The *subgradient* rule of [7], $\pi \leftarrow \pi + \alpha_k g$ with Polyak or diminishing steps, needs nothing but the last imbalance. The *bundle* rule keeps every triple $(D(\pi^{(k)}), \pi^{(k)}, g^{(k)})$ as an affine majorant of D and maximizes that polyhedral model minus a proximal term around the best price found so far, a serious step moving the center when the achieved value realizes a fraction of the predicted improvement and a null step tightening the proximal term otherwise [38]. Both rules use exactly the same messages. Because the pump commitments are integer, the running-average primal recovery of the LP model is replaced

Algorithm 1: Limited-information price coordination

Input: start price $\pi^{(0)}$: E's marginal cost of serving the metered baseline schedule (recommended), or the market price c^{buy} ; budget K ; tolerance ϵ

- 1 $k \leftarrow 0$; $\bar{C} \leftarrow +\infty$; $\underline{C} \leftarrow -\infty$
- 2 **repeat**
- 3 E $\rightarrow$ W: internal price $\pi^{(k)}$ (one scalar per pump bus and hour)
- 4 W: solve (P_W); meter the implied load $\ell^{(k)}$
- 5 W $\rightarrow$ E: load schedule $\ell^{(k)}$
- 6 E: solve (P_E); form $g^{(k)} = \ell^{(k)} - w^{(k)}$ and $D(\pi^{(k)})$; update $\underline{C}$ from the certified subproblem bounds
- 7 E: dispatch against $\ell^{(k)}$ (primal recovery); update $\bar{C}$ and the incumbent schedule
- 8 E: update $\pi^{(k+1)}$ by the subgradient or the bundle rule; $k \leftarrow k + 1$
- 9 **until** $\bar{C} - \underline{C} \leq \epsilon |\bar{C}|$ **or** the dual model predicts no improvement **or** $k = K$

Output: incumbent schedule and its certified gap; E submits the day-ahead position

by the Lagrangian heuristic standard in unit commitment [40]: at every iteration E dispatches against the schedule W has just metered, which is feasible for W by construction and, under Assumption 4, for E; its settlement is a valid upper bound on the integrated cost, and the best schedule over the iterations is the one adopted. The certified dual bound uses the solvers' proven bounds on the subproblems, so it remains valid when a subproblem stops at a tolerance.

C. What the Internal Price Is

Proposition 2 (Internal price characterization). *Fix the binaries of (34) at an integrated optimum and let $\lambda_{b,t}$ be the multiplier of the nodal balance (16) of the resulting LP. Then $\pi_{b,t}^* = \lambda_{b,t}$ is the marginal cost of serving one more kilowatt of pump load at bus b in hour t , and:*

- (i) *in an hour in which the site imports and the import rating is slack, $\pi_{b,t}^* = c_t^{\text{buy}}$; in an hour in which it exports and the export rating is slack, $\pi_{b,t}^* = c_t^{\text{sell}}$; in an hour in which the connection is idle and no PV is curtailed, $c_t^{\text{sell}} \leq \pi_{b,t}^* \leq c_t^{\text{buy}}$, the value being set by the battery's intertemporal arbitrage;*
- (ii) *a binding import rating raises $\pi_{b,t}^*$ above c_t^{buy} ; a binding export rating, or curtailed PV, lowers it below c_t^{sell} , to zero when the marginal kilowatt is served by uncurtailing;*
- (iii) *with the lossless linearization (24)–(25) and no binding voltage or branch-current limit, $\pi_{b,t}^*$ is the same at every bus; a binding feeder limit separates the buses on its two sides.*

Proposition 2 places all the informational content of coordination in the hours in which the site is not simply importing at the market price: export hours, idle hours, curtailment hours, and congested hours. When the site imports through a slack

connection, passing c_t^{buy} through to W already coordinates. Where it does not, the internal price is the site's own marginal value of energy, which no market-indexed tariff can convey.

D. Convergence and Information Footprint

Proposition 3 (Convergence, gap, and messages). *Under Assumption 4, D is concave, piecewise linear with finitely many pieces, and bounded above by $-\Pi^*$. (i) The bundle rule terminates finitely at $D^* = \max_{\pi} D(\pi)$ when the subproblems are solved exactly; with subproblems solved to a tolerance it terminates at a certified bound within that tolerance of D^* . (ii) With diminishing steps the subgradient rule converges to D^* . (iii) Every recovered schedule is feasible and its settlement is an upper bound; the incumbent is within $\bar{C} - \underline{C}$ of the integrated optimum. (iv) The duality gap $-\Pi^* - D^*$ is zero if and only if the integrated optimum remains optimal when $\mathcal{Q}$ and $\mathcal{X}$ are replaced by their convex hulls; with integer pump commitments it is generally positive, and it is the gap the operators are left with when the certificate stops improving. (v) Each iteration exchanges exactly $2|\mathcal{B}_P|T$ scalars; no water demand, tank state or limit, pump characteristic, PV forecast, battery parameter or state, feeder parameter, or market position crosses the interface.*

The gap in (iv) is the price of discreteness. It does not affect the schedule the operators adopt, which the primal recovery finds, nor the contract, which is written on the metered settlement; it affects only how tightly the operators can certify, from their own data, that no better schedule exists. The case study reports both the certified gap and the gap the central solver itself leaves.

X. IMPLEMENTATION AND EXPERIMENTAL PROTOCOL

A. Workflow

The formulation is implemented in Python with Pyomo [46], [47] as a shared model assembled from optional water and energy components, and the coordination layer on top of it (Fig. 2). EPANET input files are read through WNTR [48], and buses, loads, line parameters, and base quantities are extracted from an OpenDSS circuit [49]. A configuration file specifies the horizon, assets, prices, tariffs, connection ratings, pump-to-bus assignments, PWL segment counts, bounds, solver options, and the coordination settings, which separates the physical network description from the economic experiment. The workflow validates the configuration, runs the EPANET calibration simulation that supplies the PWL flow domains and the pump coefficients (33), and builds three kinds of model from the same components: the integrated MILP (34); the water operator's problem (P_W), which is the water block with an hourly price on the metered pump load; and the energy operator's problem (P_E), which is the energy block with a bounded, price-responsive accepted load in the bus balance, or, with that load fixed, the energy operator's response to a metered schedule. The master problem of Algorithm 1 is a small QP (bundle rule; proximal step 5×10^{-4} , doubled after a serious step and halved after a null step) or a vector update (subgradient rule with Polyak steps targeting the incumbent

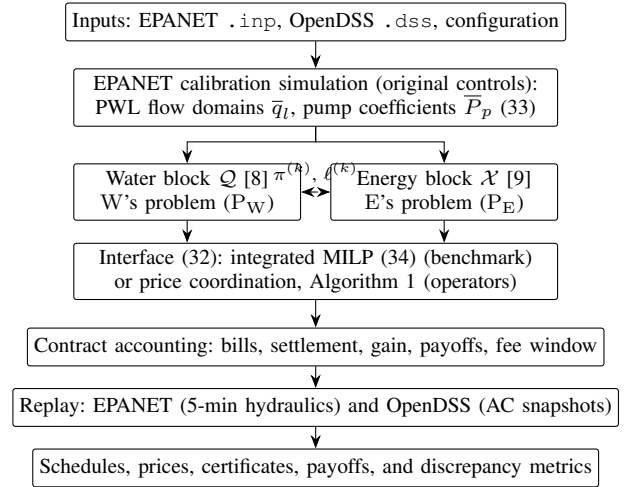


Fig. 2. Implementation workflow. The same two blocks serve the integrated benchmark and the operators' subproblems; only the internal price and the metered load schedule cross the dashed interface. MILP feasibility is separated from nonlinear replay.

primal value, as in [7]) that the energy operator solves from the load schedules it has received; prices are exchanged with a resolution of 10^{-4} unit/kWh, and each operator's subproblem is solved to a 10^{-3} relative gap or a 180 s cap, its proven bound entering the certified dual bound. All MILPs were solved with HiGHS 1.11.0 through Pyomo at a 10^{-4} relative optimality tolerance for the integrated problem, with the solver's primal-heuristic effort raised so that tightly rated instances find incumbents quickly; dependency versions and hashes of the source and input files are recorded with every run.

B. Comparison Protocol

Every arrangement is evaluated under the same networks, assets, prices, initial inventories, and terminal constraints, so that differences measure coordination and not a change of assumptions. The baseline is the two-stage flat-tariff schedule of Section VII with the energy operator responding optimally to it; the tariff benchmarks are the water operator's best responses to the pass-through and internal-price tariffs, again with the energy operator responding; the integrated optimum is the central solve. Because a tariff can leave the water operator indifferent among several schedules, the benchmark values are those of the schedule the solver returned and are reported as such. The contract is evaluated by accounting: payoffs follow from the metered settlement of each schedule and the terms (φ, F) , and the deviation test evaluates the water operator's contract payoff at every alternative schedule with the energy operator responding optimally. The distributed scheme is run with the bundle master from the energy operator's marginal-cost start and from the market-price start, and with the subgradient rule from the marginal-cost start; its recovered schedule is compared with the central solve.

C. Replay

Solver feasibility, formulation tests, and physical replay answer different questions. Unit and integration tests establish

TABLE II
CASE-STUDY DATA (SYNTHETIC PROFILES, ONE-HOUR INTERVALS)

Item	Value
Water network	SNET: 1 reservoir, 1 pump, 2 junctions, 2 pipes, 1 tank
Tank	12 m diameter, elevation 25 m, level 0–6 m, initial 3 m
Demand, pressure floor	2,214 m ³ /day with daily pattern; 20 m
Pump	50 L/s at 40 m; $\eta_p = 0.75$; $\bar{P}_p = 25.95$ kW; 6 PWL segments
Pipes, line currents	12 PWL segments (signed domain), 5 segments
Feeder	Two-bus circuit, load shape peaking at 100 kW
PV, battery	200 kW _p ; 200 kWh, 50 kW, $\eta = 0.95$, $e^0 = 100$ kWh
Connection	150 kW import, 50 kW export; $0.9 \leq v \leq 1.1$ p.u.
Prices (unit/kWh)	purchase 0.08/0.16/0.32 (off/mid/on-peak); sale 0.03; flat tariff 0.16

that the constraints conserve inventory, enforce asset limits, preserve units, route the pump load to the correct bus, and that the contract accounting identities hold. Algebraic residuals of the solved MILPs establish consistency with the implemented surrogate. Replaying the optimized decisions in independent nonlinear simulators assesses physical fidelity on the instance: the pump statuses are imposed on the original EPANET network with a five-minute hydraulic time step and hourly reports, and the optimized net bus injections are solved as balanced constant-power snapshots in OpenDSS.

XI. CASE STUDY

A. Test System and Data

The water operator runs the SNET network of the repository: one ground-level reservoir, one fixed-speed pump (design point 50 L/s at 40 m, wire-to-water efficiency 0.75, calibrated on-state power $\bar{P}_p = 25.95$ kW), two junctions, two pipes, and one elevated cylindrical tank (12 m diameter, 25 m elevation, level 0–6 m, initial 3 m). Daily demand is 2,214 m³ with the diurnal pattern of Fig. 3; a 20 m pressure floor and terminal replenishment are enforced. The tank holds about 2.8 pump-hours of water, so the shiftable volume is tank-limited in the sense of the two-period model. The energy operator owns the two-bus feeder of the repository with its daily load shape scaled to a 100 kW peak, a 200 kW_p PV array, a 200 kWh/50 kW battery ($\eta = 0.95$, initial and minimum terminal energy 100 kWh), and a connection rated 150 kW import and 50 kW export. Day-ahead purchase prices are 0.08 in hours 0, 1, 2, 3, 4, 5, 6, 21, 22, and 23; 0.16 in hours 7, 8, 9, 10, 11, 12, 13, 14, and 15; 0.32 in hours 16, 17, 18, 19, and 20 (unit/kWh); the sale price is 0.03. The flat retail tariff of the baseline is $\bar{\tau} = 0.16$, the time average of the purchase price. Profiles are synthetic and illustrative (Table II).

B. Experiments

Six experiments are run: (E1) the flat-tariff baseline and the two tariff benchmarks against the integrated optimum; (E2) the payoffs of the revenue-sharing contract, the fee window, and a numerical deviation test; (E3) the replay of the

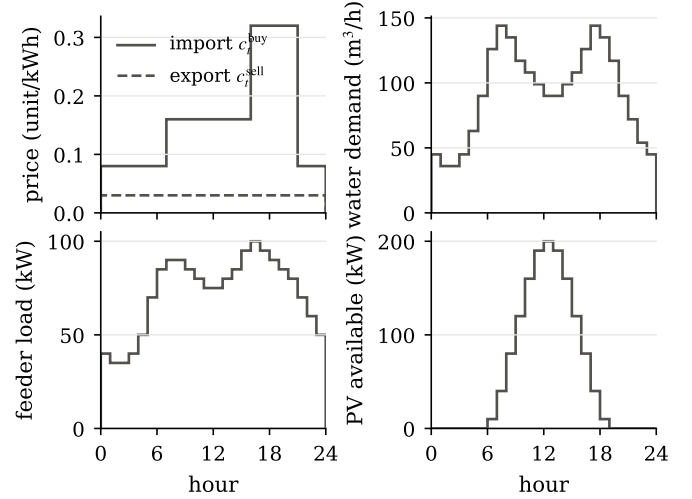


Fig. 3. Input profiles: day-ahead purchase and sale prices, water demand, feeder load, and available PV power.

baseline and coordinated schedules in EPANET and OpenDSS; (E4) the price coordination scheme with the bundle master started from the energy operator's marginal costs and from the market price, and with the subgradient rule of the two-agent model; (E5) sweeps of the import and export ratings of the connection; and (E6) the restricted-LP internal prices that verify Proposition 2. All models were solved with HiGHS 1.11.0 through Pyomo; the integrated MILP has 3219 variables and 2598 constraints and was solved to a 0.003% relative gap in 76 s, the coordination subproblems to a 0.1% gap. Timing includes the Pyomo interface overhead and is not a solver benchmark.

C. The Value of Coordination (E1, E6)

Table III compares the arrangements. Under the flat tariff the water operator needs 9 pump-hours and, holding its level, places 2 h at 0.08, 5 h at 0.16, 2 h at 0.32 (unit/kWh); the site's day-ahead cost is 107.94/day. The integrated optimum pumps 3 h at 0.08, 6 h at 0.16, 0 h at 0.32 and costs 90.91/day, a coordination gain $\mathcal{G} = 17.03$ /day, 15.8% of the site's bill and 46% of the water operator's baseline pumping bill of 37.37/day. Figure 4 shows how the gain is earned: the coordinated schedule places 5 of its 9 pump-hours in the export and idle hours, where the site's own energy is cheapest, and none in the peak block at 0.32; the baseline, holding its level, places 4 in the surplus hours and 2 in the peak block, which the grid serves at the peak price.

The internal price π_t^* of the restricted LP follows Proposition 2: in the 15 hours in which the site imports it equals the purchase price; in the 6 export hours (hours 9, 10, 11, 12, 13, and 14) it equals the sale price 0.03; in the 3 idle hours (hours 7, 8, and 16) it lies inside the band $[c_t^{\text{sell}}, c_t^{\text{buy}}]$, set by the battery's intertemporal arbitrage. Passing the purchase price through to the water operator captures 95.6% of the gain: the water operator then avoids the peak block but places only 5 pump-hours in the surplus hours, having no reason to prefer them over the other hours of the same purchase price. Charging

TABLE III
ARRANGEMENTS COMPARED (UNIT/DAY). GAIN AND CAPTURE ARE
RELATIVE TO THE BASELINE

Arrangement	Cost	Gain	Capt. (%)	On (h)	Import	Export
Baseline: flat tariff, level holding	107.94	0.00	0.0	9	856	212
Pass-through tariff $\tau_t = c_t^{\text{buy}}$	91.66	16.28	95.6	9	830	201
Internal-price tariff $\tau_t = \pi_t^*$	91.21	16.73	98.2	9	830	216
Integrated optimum (contract)	90.91	17.03	100.0	9	830	226

TABLE IV
DAILY PAYOFFS BY ARRANGEMENT (UNIT/DAY); FEE ANCHORED ON
THE BASELINE

Arrangement	φ	F	Π_W	Π_E	Π	$\Delta\Pi_W$	$\Delta\Pi_E$
Baseline (flat tariff)	–	–	-37.37	-70.57	-107.94	–	–
Contract	0.00	37.37	-37.37	-53.54	-90.91	+0.00	+17.03
Contract	0.50	-16.60	-28.85	-62.05	-90.91	+8.51	+8.51
Contract	1.00	-70.57	-20.34	-70.57	-90.91	+17.03	+0.00

TABLE V
WATER OPERATOR'S CONTRACT PAYOFF AT $\varphi = 1/2$ FOR ALTERNATIVE
SCHEDULES (UNIT/DAY)

Schedule run by W	Π_W	Deviation gain
Baseline: flat tariff, level holding	-37.37	-8.51
Pass-through tariff $\tau_t = c_t^{\text{buy}}$	-29.23	-0.38
Internal-price tariff $\tau_t = \pi_t^*$	-29.00	-0.15
Integrated optimum (contract)	-28.85	+0.00

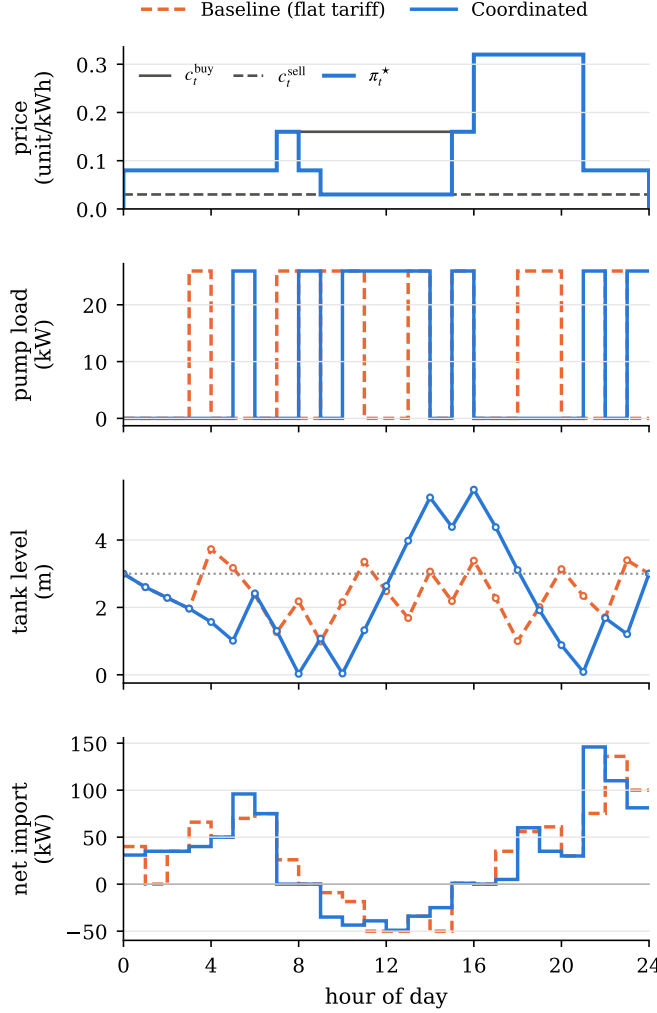


Fig. 4. Baseline versus coordinated operation. Top to bottom: purchase and sale prices with the internal price π_t^* ; pump load; tank level (markers are inventory boundaries, the dotted line the initial level); net import at the connection (negative is export).

the internal price itself captures 98.2%: the water operator is indifferent among equally priced hours, and the tie it happened to break is not the one the energy operator can serve most cheaply (Proposition 1).

D. Dividing the Gain (E2)

Table IV lists the payoffs. Because the site is a net importer, the settlement is negative ($R^* = -90.91/\text{day}$), and the contract is read as bill sharing: with the fee anchored on the baseline, $F(\varphi) = \text{bill}^0 + \varphi R^0$, the water operator's payoff rises by $\varphi\mathcal{G}$ and the energy operator's by $(1 - \varphi)\mathcal{G}$ at every share

(Corollary 1). At the equal split $\varphi = 1/2$ the fee is -16.60/day and each operator gains 8.51/day; for that share any fee in $[-25.11, -8.09]$ leaves both operators no worse off than the baseline, an interval of width $\mathcal{G}$. The deviation test in Table V evaluates the water operator's contract payoff at $\varphi = 1/2$ if it ran one of the other schedules instead, with the energy operator responding optimally: every deviation loses, as Theorem 1 requires.

E. Distributed Computation (E4)

Table VI and Fig. 5 summarize the price coordination runs. Started from the energy operator's marginal cost of serving the metered baseline schedule, the first message of the recommended protocol, the scheme certifies a gap of 0.72% after the first exchange, the recovered schedule attains the integrated optimum at iteration 2, and the bundle master stops after 7 iterations, when its model predicts no further improvement at the exchanged price resolution, with a dual bound of 90.55/day and a certified gap of 0.39%. Started from the market price instead, the first exchange certifies only 8.18%, because the purchase price leaves the energy operator indifferent in every import hour; the scheme attains the optimum at iteration 7, reaches a 1% certificate at iteration 17, and ends after 29 iterations with a gap of 0.02%. The subgradient rule from the marginal-cost start attains the optimum at iteration 2 and reaches a gap of 0.39% after 15 iterations. Evaluating the dual function at the restricted-LP prices π^* gives $D(\pi^*) = 90.91/\text{day}$ against $C^* = 90.91/\text{day}$: the Lagrangian relaxation of the interface has no duality gap on this instance (Proposition 3(iv)), so the residual certified gap of the runs is the price of finite iterations, quantized prices, and subproblem tolerances, not a structural limit. For comparison, the central solver's own bound on the integrated MILP is 90.90, a gap of 0.003%, and it holds both operators' data. Each iteration exchanges 48 scalars; no demand forecast, tank state, pump characteristic, PV forecast, battery state, or market position crosses the boundary.

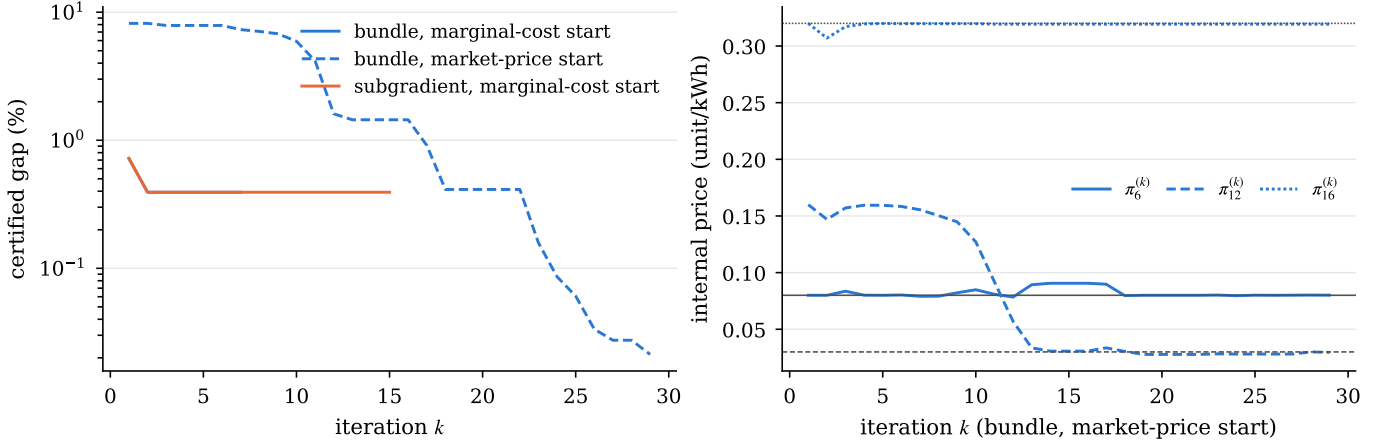


Fig. 5. Left: certified gap between the recovered primal value and the certified dual bound per iteration for the bundle master from the marginal-cost and market-price starts and for the subgradient rule. Right: internal-price iterates at three hours for the bundle run from the market-price start; thin lines mark the restricted-LP prices π_t^* .

TABLE VI

PRICE COORDINATION RUNS: ITERATIONS, MESSAGES, CERTIFIED DUAL AND PRIMAL VALUES, FINAL GAP, THE ITERATION AT WHICH THE RECOVERED SCHEDULE FIRST ATTAINS THE INTEGRATED OPTIMUM, AND WALL TIME

Run	Iter.	Msgs	Dual	Primal	Gap %	Opt. at	Min
Bundle, marginal-cost start	7	336	90.55	90.91	0.39	2	4.5
Bundle, market-price start	29	1392	90.89	90.91	0.02	7	57.2
Subgradient, marginal-cost start	15	720	90.55	90.91	0.39	2	10.7

F. Connection-Limit Sensitivities (E5)

Table VII and Fig. 6 vary the ratings of the connection. Tightening the import rating lowers the gain from 17.03/day at 150 kW to 10.92/day at 70 kW; at 100 kW and below the internal price departs from the market price in hours 21, 22, and 23, where the rating binds. No schedule is feasible at 60 kW. Tightening the export rating lowers the gain from 17.03/day at 50 kW to 16.61/day at 0 kW: with less room to export, both schedules absorb more of the midday surplus on site, and the internal price in the curtailment hours falls to the marginal value of spilled energy, zero, for the coordinated schedule and the baseline alike.

G. Independent Nonlinear Replay (E3)

The baseline and coordinated pump schedules are replayed in EPANET with a five-minute hydraulic time step and hourly reports, the level controls of the input file being replaced by the hourly commands, and the optimized net bus injections are solved as balanced constant-power snapshots in OpenDSS. Table VIII summarizes the discrepancies. Both schedules satisfy the pressure floor and the voltage band at every report time and the imposed pump statuses match the MILP decisions. The replay ends with the tank 0.137 m (baseline) and 0.124 m (coordinated) below its initial level despite the replenishment constraint (9), so the hourly hydraulic approximation does not preserve the terminal requirement exactly. The electrical discrepancies are small because the feeder is short and lightly loaded, the regime in which the flat-voltage linearization is

TABLE VII

CONNECTION-LIMIT SWEEPS (UNIT/DAY). "OFF REFERENCE" LISTS THE NON-IDLE HOURS IN WHICH THE INTERNAL PRICE DEPARTS FROM THE MARKET PRICE OF THE CONNECTION'S DIRECTION

Limit	kW	Baseline	Integrated	Gain	Off reference
Import	150	107.94	90.91	17.03	none
Import	125	107.94	90.91	17.03	none
Import	100	110.28	98.67	11.61	hours 21, 22, and 23
Import	90	116.54	105.12	11.43	hours 21, 22, and 23
Import	80	122.99	111.75	11.24	hours 5, 21, 22, and 23
Import	70	129.57	118.64	10.92	hours 5, 6, 21, 22, and 23
Import	60	infeasible	infeasible	—	—
Export	50	107.94	90.91	17.03	none
Export	25	110.26	93.18	17.09	hours 9, 10, 11, 12, 13, and 14
Export	0	114.29	97.68	16.61	none

TABLE VIII

REPLAY OF THE BASELINE AND COORDINATED SCHEDULES IN EPANET AND OPENDSS

Metric	Baseline	Coordinated
Minimum junction pressure (m)	25.92	25.03
Pressure violations at report times	0	0
Pump-status mismatches at report times	0	0
Maximum tank-head discrepancy (m)	0.139	0.125
Maximum flow discrepancy (L/s)	0.350	0.374
Terminal tank-head change vs. initial (m)	−0.137	−0.124
All OpenDSS snapshots converged	yes	yes
Maximum voltage discrepancy (p.u.)	0.00002	0.00003
Voltage violations	0	0
Peak line loading (%)	0.17	0.19
Maximum grid-exchange discrepancy (kW)	0.000	0.000

most accurate. Report-time checks do not certify feasibility between reports; the results support the schedules as planning-stage outputs, not as a certification for field operation.

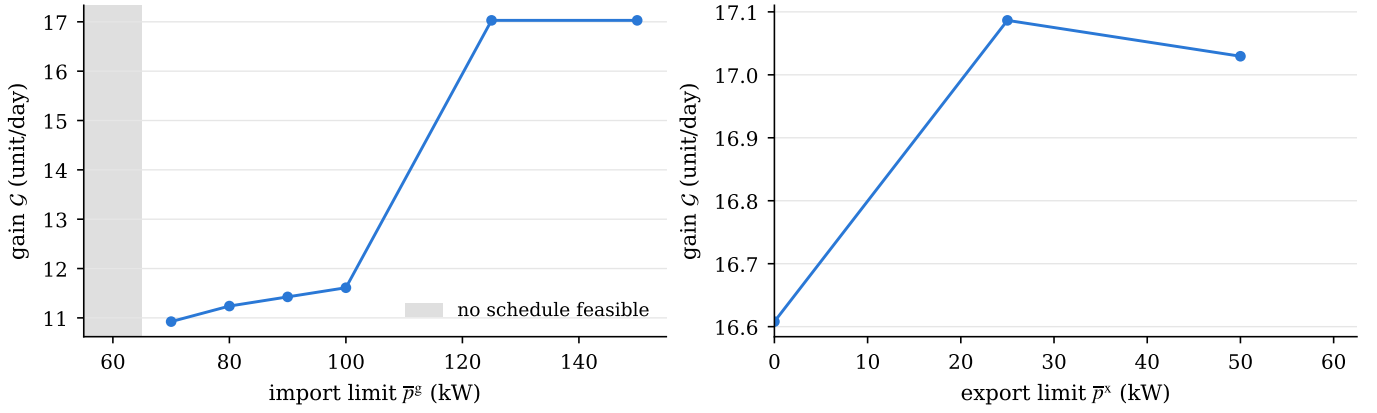


Fig. 6. Coordination gain versus the import rating (left) and the export rating (right) of the connection. Shaded bands mark ratings at which only coordinated operation is feasible or no schedule is feasible.

XII. DISCUSSION

A. What Generalizes

Four points carry beyond the instance. *Tariffs select schedules; they do not coordinate.* A flat tariff fixes the water operator's pump-hours and leaves their timing to habit; a pass-through tariff aligns timing with the market but not with the site, and it captured 95.6% of the gain here only because the site imports at the market price in most hours; charging the internal price itself captured 98.2% because the on/off pump makes ties between equally priced hours generic, and a tariff cannot break them in the site's favor. *Share the settlement, not audited costs.* Because the water operator's controllable costs are fixed by demand up to the granularity of its pumps, a share of the verifiable market settlement plus a fixed fee aligns incentives exactly, with no open books and no cost-allocation dispute; for a net importer the same contract is a share of the site's bill with a rebate. *The right split is a fee, not a share.* Anchoring the fee on the baseline turns any share into a proportional split of the gain, so bargaining is over one number, and the equal split has a focal point. *Coordinate the plan, contract the outcome.* An internal price and a metered schedule, exchanged a few dozen times, compute the jointly optimal schedule and certify how far from optimal it can be; the contract makes the result self-enforcing. Privacy costs $2|\mathcal{B}_P|T$ numbers per message.

B. Discreteness

The two-agent LP model of [7] had no duality gap and a unique best response to the internal price. Both properties fail with on/off pumps, and the case study shows what replaces them. The recovered schedule reaches the integrated optimum within a handful of exchanges, and the certificate the operators can compute from their own data ends at a gap of 0.39% from the marginal-cost start. On this instance the dual function evaluated at the restricted-LP prices lies 0.000% below the integrated cost, so the Lagrangian relaxation has essentially no duality gap and the residual is the price of finite exchanges, quantized prices, and subproblem tolerances; in general the integer commitments can leave a gap that no price closes

(Proposition 3), and the central solver, holding all the data, itself leaves a gap of 0.003%. And the internal price no longer selects a unique schedule: at π^* the water operator is indifferent between several hours of equal price, and it is the energy operator's response to the metered schedule, inside the planning loop, that picks the one the site can serve most cheaply. These are the two places where the planning protocol does work that a price alone cannot do, and they are the reason the protocol is part of coordinated operation rather than a computational convenience.

C. Limitations

The fixed on-state pump coefficient (33) is the principal coupling approximation. It keeps the interface linear and metered, but it does not reward operating a pump near its best-efficiency point, and the modeled pump energy is only as accurate as the calibration trajectory; a validated PWL relation between electrical power and the hydraulic operating point is the natural next step, and it would keep the interface linear in the metered load. The hourly discretization can leave the replayed tank slightly below its terminal requirement (Table VIII); a reserve margin or a simulation-informed correction of (7) would close the gap, and all compared schedules would have to share it. A small synthetic water network and a two-bus feeder do not establish scalability, behavior under feeder congestion, or performance on real operator data; the case does show connection congestion, which is where the internal price departs from the market price. The linearized power flow is most defensible near its expansion point. Water quality, leakage, battery wear, pump switching costs, equipment outages, and forecast uncertainty are absent. The contract analysis is deterministic: under uncertainty the transfer shares whatever settlement materializes, and the allocation of risk between the operators becomes a design question in its own right (Remark 2).

D. Implementation Prerequisites

The contract needs only settlement-grade artifacts, the market settlement statement and the pump submeter, but it does

need a legal wrapper: a revenue-sharing agreement with (φ, F) negotiated once and revisited periodically, and, for a regulated utility, its regulator's approval of market-indexed income. Corollary 1 bounds where the terms can be, and the equal split gives a focal point; the outcome will reflect bargaining power and outside options. The planning protocol needs an hourly price and load exchange the day before delivery, which existing demand-response interfaces already provide, and it needs each operator to run its own scheduling problem, which the water operator already does under any hourly tariff. None of these barriers is analytic, but they bound where the scheme applies, and the modest scale of the gain argues for keeping the contractual machinery as light as (φ, F) allow.

XIII. CONCLUSION

This paper modeled the day-ahead coordination of two independently managed local operators, a water utility with schedulable pumping and storage and an energy operator whose PV, battery, feeder, and market interface can monetize that flexibility, as a contracting problem while keeping the physics of both networks. The water operator's decisions were constrained by the MILPNet formulation and the energy operator's by the electrical part of the DES model of Morvaj *et al.*; the two met in one metered quantity, the pump load at its supply bus, and their joint optimum was a MILP. Flat, pass-through, and internal-price tariffs proved unable to coordinate: the first fixes only pump-hours, the second the wrong price in export and congested hours, and the third leaves ties and dictates the division. A revenue-sharing contract with a fixed fee coordinated at every share, its proof unaffected by the integer commitments, and anchoring the fee on the baseline bill turned the share into a proportional split of the gain, which for a net-importing site reads as bill sharing with a rebate. A Lagrangian relaxation of the interface let the operators compute the schedule by exchanging only an internal price and a metered load schedule; a bundle master certified the dual bound where subgradient steps could not, and the price was characterized as the site's nodal marginal cost, equal to the purchase price in import hours, the sale price in export hours, and different from both only where the connection or the feeder is at a limit. In the case study coordination was worth 17.03 per day, 46% of the pumping bill; the pass-through tariff captured 95.6% of it, the contract all of it at every share, and the distributed scheme found the schedule within a few exchanges from the energy operator's own marginal costs while certifying a gap of 0.39%.

Three extensions follow naturally. First, uncertainty: bidding under PV and price uncertainty with imbalance settlement turns the integrated problem into a two-stage stochastic program; the contract's affine structure extends scenario by scenario, but risk allocation between the operators becomes a design question. Second, a flow- and head-dependent pump power model and larger networks with several pump groups and feeder congestion, which enrich $\mathcal{Q}$ and $\mathcal{X}$ and leave the architecture intact. Third, one-to-many coordination: an energy operator contracting simultaneously with water, thermal, and EV-charging flexibility providers, where the hierarchical

decompositions of [35] apply and the contracting layer raises classical questions of rent division among multiple flexibility suppliers.

APPENDIX PROOFS

Throughout, $\Delta t = 1$ to lighten notation, $C = -R$ denotes the site's day-ahead cost, $C^* = -\Pi^*$, and $C_E(\ell) = -R_E(\ell)$ is the energy operator's least cost of serving a metered load ℓ .

A. Proof of Lemma 1

Summing (1) over $n \in \mathcal{J}$ and $t \in \mathcal{T}$ and (7) over $k \in \mathcal{K}$ and $t \in \mathcal{T}$, every link flow that joins two junctions appears once with each sign and cancels; what remains is the flow leaving the reservoirs, the demand, and the net tank inflow, which telescopes to $A_k(h_{k,T} - h_k^0)/\Delta t_s$. Rearranging gives the identity. Under (9) the storage term is nonnegative. At a schedule that no positive electricity price improves, no pump-hour can be removed while keeping the schedule feasible; since removing pump-hours is the only way to reduce the storage term, that term is at its least feasible value for the given number of pump-hours, which is the granularity statement. Pumping energy is $\sum_p \bar{P}_p \sum_t u_{p,t}$, and two feasible schedules can differ in $\sum_t u_{p,t}$ because the flow delivered in an on-hour is the curve flow at the head the pump faces, which depends on the tank level and the network state in that hour. ■

B. Proof of Proposition 1

(i) With $\tau_t \equiv \bar{\tau}$ the objective of (35) is $\bar{\tau} \sum_p \bar{P}_p \sum_t u_{p,t}$, a function of the pump-hour counts alone; every schedule with the minimal counts attains it, so all are best responses and the tariff does not distinguish their timing. If u^* has more pump-hours than the minimum, its bill is strictly larger than the minimum under every $\bar{\tau} > 0$, so it is not a best response. (ii) Under $\tau_t = c_t^{\text{buy}} + \eta$ the objective is $\sum_t c_t^{\text{buy}} \ell_t(u) + \eta \sum_t \ell_t(u)$; on the set of schedules with a given pump-hour count the second term is constant, and across counts it only adds a constant per hour, so best responses minimize $\sum_t c_t^{\text{buy}} \ell_t$ up to that constant. By Proposition 2, $\pi_t^* \leq c_t^{\text{buy}}$ with strict inequality in export, idle, and curtailment hours; if u^* pumps in such an hour, moving that pump-hour to any hour with c_t^{buy} no larger and π_t^* strictly larger, when feasible, does not increase the pass-through bill but increases the internal-price bill, so u^* need not minimize the former, and it fails to whenever such a move is feasible; in the case study it does. Indifference between hours of equal price is immediate. (iii) If the Lagrangian relaxation of (32) has no duality gap, (u^*, ξ^*) minimizes the Lagrangian at π^* , so u^* minimizes $\sum_t \pi_t^* \ell_t(u)$ over $\mathcal{Q}$ and is a best response to $\pi^* + \eta$; any schedule with the same price-weighted load is also a best response, hence ties. Under $\tau = \pi^* + \eta$ the water operator's payoff is $-\sum_t (\pi_t^* + \eta) \ell_t^*$, affine and strictly decreasing in η , and the energy operator's is C^* less that bill; the attainable divisions of $\mathcal{G}$ form the half-open interval of payoffs so generated, which is bounded above by the baseline participation constraint only for η in a range that need not intersect the range required by the other operator's constraint. ■

C. Proof of Theorem 1

E's side. Given u^* , E's payoff is $(1-\varphi)R(\xi) + F$ with $\xi \in \mathcal{X}(\ell(u^*))$. For $\varphi \leq 1$ this is a nondecreasing transformation of R , and ξ^* maximizes R over $\mathcal{X}(\ell(u^*))$ by block optimality of the integrated optimum; hence no profitable deviation exists. *W's side.* Given ξ^* , a deviation $u' \in \mathcal{Q}$ is available only if the operating point obtained by fixing E's decisions and letting the metered exchange adjust to $\ell(u')$ remains feasible; on that set the site's settlement is R itself evaluated at a point feasible for (34), so $R(u', \xi') \leq \Pi^* = R(u^*, \xi^*)$. W's payoff change is $\varphi[R(u', \xi') - R(u^*, \xi^*)] \leq 0$ for every $\varphi \geq 0$, so no profitable deviation exists, proving (i). For (ii), transfers cancel in the sum, $\Pi_W + \Pi_E = \Pi^*$; for any target division $(a, \Pi^* - a)$ choose any $\varphi \in [0, 1]$ and $F = \varphi\Pi^* - a$. Nothing in the argument uses convexity of $\mathcal{Q}$ or $\mathcal{X}$. ■

D. Proof of Corollary 1

W is no worse off than in the baseline iff $\varphi R^* - F \geq -\text{bill}^0$, i.e., $F \leq \text{bill}^0 + \varphi R^*$; E iff $(1-\varphi)R^* + F \geq R^0 + \text{bill}^0$, i.e., $F \geq \text{bill}^0 + R^0 - (1-\varphi)R^*$. The interval is nonempty because its width is $\varphi R^* + (1-\varphi)R^* - R^0 = \mathcal{G} \geq 0$. Substituting $F = \text{bill}^0 + \varphi R^0$ gives $\Pi_W = \varphi(R^* - R^0) - \text{bill}^0 = -\text{bill}^0 + \varphi\mathcal{G}$ and $\Pi_E = R^* - \varphi R^* + \text{bill}^0 + \varphi R^0 = R^0 + \text{bill}^0 + (1-\varphi)\mathcal{G}$; the anchored fee lies in the window because $R^0 \leq R^*$. With fallbacks at the baseline payoffs, the Nash bargaining solution maximizes $(\varphi\mathcal{G})((1-\varphi)\mathcal{G})$, attained at $\varphi = 1/2$. ■

E. Proof of Proposition 2

With the binaries fixed, (34) is an LP in which the pump load enters only the right-hand side of the nodal balance (16) at the pump buses; by LP sensitivity its multiplier $\lambda_{b,t}$ is the marginal cost of an additional kilowatt of load there, and it is also the multiplier of the relaxed interface (32), which is the internal price. (i) At a PCC bus with a slack import rating and $y^g = 1$, stationarity in $p_{b,t}^g$ gives $\lambda_{b,t} = c_{b,t}^{\text{buy}}$ when the import is strictly positive (its lower bound is slack); symmetrically $\lambda_{b,t} = c_{b,t}^{\text{sell}}$ in an export hour with a slack export rating. In an idle hour both variables sit at their lower bound, so stationarity yields $c_{b,t}^{\text{sell}} \leq \lambda_{b,t} \leq c_{b,t}^{\text{buy}}$; the value within the band is determined by the stationarity conditions of the battery variables that are strictly between their bounds, i.e., by the intertemporal arbitrage. (ii) A binding import rating adds its nonnegative multiplier to $c_{b,t}^{\text{buy}}$; a binding export rating subtracts one from $c_{b,t}^{\text{sell}}$. If PV is curtailed in hour t , $p_{b,t}^{\text{pv}}$ is strictly inside its bounds, and stationarity in it gives $\lambda_{b,t} = 0$ at its bus. (iii) The lossless linearized flows make the nodal balances of a connected feeder sum to the PCC balance with the line flows cancelling; with no voltage or current limit binding, the line-flow, voltage, and angle variables are unconstrained beyond (24)–(25), and stationarity in them forces all $\lambda_{b,t}$ to coincide; a binding limit on a line introduces its multiplier as the difference between the two end buses. ■

F. Proof of Proposition 3

D is a minimum of finitely many affine functions of π , one for each pair of an integer assignment in $\mathcal{Q}$ and a vertex

of the corresponding polyhedron in $\mathcal{X}$, hence concave and piecewise linear; weak duality gives $D(\pi) \leq C^*$ for every π . (i) Each cut is an affine majorant of D that is exact at the price where it was generated; the bundle rule with serious and null steps on a polyhedral concave function terminates finitely at a maximizer [38], and inexact subproblem values shift each cut by at most the subproblem tolerance, so the certified bound, computed from the solvers' proven bounds, lies within that tolerance of D^* . (ii) is the standard convergence of the projected subgradient method with diminishing steps for concave functions with bounded supergradients [39]. (iii) The metered schedule is feasible for W by construction and the response is feasible for E by Assumption 4, so their value is the cost of a feasible point of (34), and the incumbent is within the certified gap of C^* . (iv) By the Geoffrion characterization, D^* equals the value of (34) with $\mathcal{Q}$ and $\mathcal{X}$ replaced by their convex hulls, so the gap vanishes iff the integer optimum is also optimal for that convexified problem. (v) The message count is read off Algorithm 1: one vector $\pi^{(k)} \in \mathbb{R}^{|\mathcal{B}_P|T}$ and one vector $\ell^{(k)} \in \mathbb{R}^{|\mathcal{B}_P|T}$ per iteration; the master problem is built from those vectors and E's own subproblems. ■

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DATA AND CODE AVAILABILITY

Network inputs, configuration, optimization and coordination code, tests, and the experiment scripts that regenerate Section XI and its figures are included in the accompanying repository. From the repository root, `python -m src.experiments` regenerates the study from `data/inputs/paper_case.yaml`, and `latexmk -pdf main.tex` in the paper directory rebuilds the manuscript. The bundled MILPNet reference code is attributed to Thomas and Sela [8] and is distinct from the project implementation. No confidential measurements or external market data are required.

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